

- JUAN AFANADOR, *The leftmost sinister paraconsistency*.

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This work formalises Paraconsistent Mode Theory [1] within the BiSikkel framework [2], leveraging Multimode Type Theory to internalise paraconsistency and intuitionistic reasoning as two distinct computational modes. I model the relationship between **Act** (intuitionistic) and **Para** (paraconsistent) modes via an essential geometric morphism between 2-toposes. These modes I bridge using a sinister morphism μ_{sin} [3] identified with the geometric morphism's leftmost adjoint. Instead of confining contradictions behind the fibrational lock of a negative modality, this identification induces a positive modality acting as an opfibrational gate.

That is, the proposed structure preserves colimits, rendering contradictions accessible and dynamic events. Semantically grounded in a 2-classifier of discrete opfibrations [4], these properties further validate a non-trivial boundary operator ∂A . And so I can show that the logical witnesses to these boundaries are *reified* into first-class terms $\Gamma \vdash_{\text{Act}} \text{event}(p) : \Sigma_{\mu_{\text{sin}}}(\partial A)$, transmuting the semantic fact of contradiction into a syntactic resource. Logical consistency in **Act** is preserved by formalising transparent morphisms as parametric right adjoints, serving as controlled interfaces for pattern-matching on contradictions qua modal events.

[1] JUAN AFANADOR, *Paraconsistency in a Mode Theory*, **Book of Abstracts of the 7th World Congress on Paraconsistency** (Oaxaca, Mexico), (Luis Estrada-González and María del Rosario Martínez-Ordaz, editors), Universidad Benito Juárez de Oaxaca, 2024, pp. 28–31

[2] JORIS CEULEMANS, ANDREAS NUYTS, AND DOMINIQUE DEVRIESE, *BiSikkel: A Multimode Logical Framework in Agda*, **Proceedings of the ACM on Programming Languages**, vol. 9 (2025), pp. 210–240.

[3] MICHAEL SHULMAN, *Semantics of Multimodal Adjoint Type Theory*, **Electronic Notes in Theoretical Informatics and Computer Science**, vol. 3 (2023), pp. 18:1–18:19.

[4] LUIS ESTRADA-GONZÁLEZ, *The Evil Twin: The Basics of Complement-Toposes*, **Springer Proceedings in Mathematics & Statistics**, vol. 152 (2015), pp. 375–425.